# Dennis Zubov

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## Education

**Georgia Institute of Technology | Atlanta, GA** *August 2020 – Present*
Bachelor of Science in Computer Engineering (4.0/4.0 GPA) with Minor in Economics Expected Graduation, May 2024

- **Concentrations**: Distributed System & Software Design, Information Internetworks

**Georgia Tech Lorraine | Metz, France** *January – May 2022*

- Completed 14 credit hours of coursework (4.0/4.0 GPA)

## Skills

**Programming:** Python, Java, C, C++, SQL, HTML5, CSS, Linux, OS X, JavaScript, TypeScript, VHDL, Swift, MATLAB, Assembly
**Web Frameworks & Technologies:** Django, Flask, Angular.js, Git, GitLab, GitHub, Jira, Unit Testing, REST API, CI/CD, React, MySQL, PostgreSQL, MySQL, AWS, DynamoDB, MongoDB, Bootstrap
**Communication:** Presentations (Large and Small Audiences), Technical Reports
**Languages:** English (native), French (conversational), Russian (fluent)

## Experience

**Jurumani Solutions | Johannesburg, South Africa** *February – July 2020*
***Software Engineering Intern***

- Built more than two hundred backend HTTP API endpoints in Python that were deployed to production. Used the Django web framework to act as an API gateway for a web application for a large mobile services provider.
- Expanded the functionality and coverage of the codebase's CI/CD that was hosted on Gitlab. Ensured that all the code that I pushed had complete test coverage and created tests for older API endpoints that did not have test coverage.
- Collaborated with a software engineering team of around 10 engineers to build a large full stack web application.
- Used agile development methodology with daily standups and weekly meetings with our client to ensure that the client's wants and needs were being met. Used Jira to keep track of the progress on our 2-week long sprints.
- Tested the API endpoints by building hundreds of unit tests using the Insomnia testing framework ensuring that all common and edge cases were covered.

## Projects

**Music Blog** *2022*
*Created a music blog that allows users to make blog posts and upload music.*

- Built a full-stack website using the Django framework to manage backend communications with the relational database and to display dynamic web pages for the end user. Built using Python, JavaScript, HTML, and CSS.
- Developed full login functionality with profile management including functionality such as uploading a profile picture.
- Implemented a SQL database to store blog and user information. Managed the versions of my website using Git to closely track the different phases of the project. The code base is currently hosted on my GitHub account.

**Photo Gallery** *2022*
*Created a photo gallery website that allows users to post, delete, and update photos within photo albums.*

- Built website using AWS services such as S3 buckets, DynamoDB, RDS, and EC2. Created public photo gallery with full login functionality with email confirmation, session management, password hashing. Built in Python, using Flask.
- Implemented SQL and NoSQL version of this photo gallery with CRUD operations for photos and albums. Stored the raw image files in S3 buckets. Used DynamoDB and MySQL to store data related to photo, albums, and users.
- Created an AWS EC2 instance to host and run the codebase. Implemented multiple AWS services into my website including SES, IAM, S3, EC2, RDS, and DynamoDB.

**RPG Quest Video Game** *2021*
*Created a video game where the player must interact with characters and objects from multiple maps.*

- Built a top-down video game using four action buttons on a bread board with an accelerometer for character movement. Using an Mbed Microcontroller to connect components. Implemented custom hash table for my C codebase.
- Utilized an LCD screen to display the game. The player interacts with characters and objects to get enough information to travel to the dragon's lair to fight the dragon and retrieve the key which unlocks the final door.

## Relevant Coursework

**Software:** Data Structures, Algorithms, Mutex Locks, Multithreading, Agile Web Development, Computer Networking**,** Relational Databases, NoSQL Databases, Cloud Computing
**Hardware:** FPGA, Oscilloscopes, Logic Analyzers, Waveform Generators, Mbed Microcontroller
**Mathematics:** Probability, Statistics, Linear Algebra, Multivariable Calculus, Differential Equations, Combinatorics
**Finance:** Advanced Macroeconomics, Money & Capital Markets

## Leadership & Other Experience

**Georgia Institute of Technology | Atlanta, GA** *February 2021 – Present*
***Intramural Supervisor (~12 hours per week)***

- Managing the logistics of over a dozen intramural sports that are offered to Georgia Tech students.
- Leading flag football and dodgeball training programs for new officials including classroom and on-field/court trainings multiple times a semester. Teaching classroom session using slideshows that we create, maintain, and update as needed.
- Responding to injuries and managing any conflict during sports games. Active Red Cross First Aid and CPR certification.

**Georgia Tech Investments Committee | Atlanta, GA** *August 2022 – Present*
***Quantitative Sector Analyst (~6 hours per week)***

- Analyze financial markets from multiple different sectors to evaluate the potential gains or losses from a security.
- Manage and invest an endowment of approximately $2 million collectively with the Investments Committee.
- Learn accounting and finance concepts to effectively analyze financial markets and create detailed pitches.
- Developing python programs/tools that aid in the investment decisions of the committee through quantitative analysis.